



Electrical Submersible Pumps

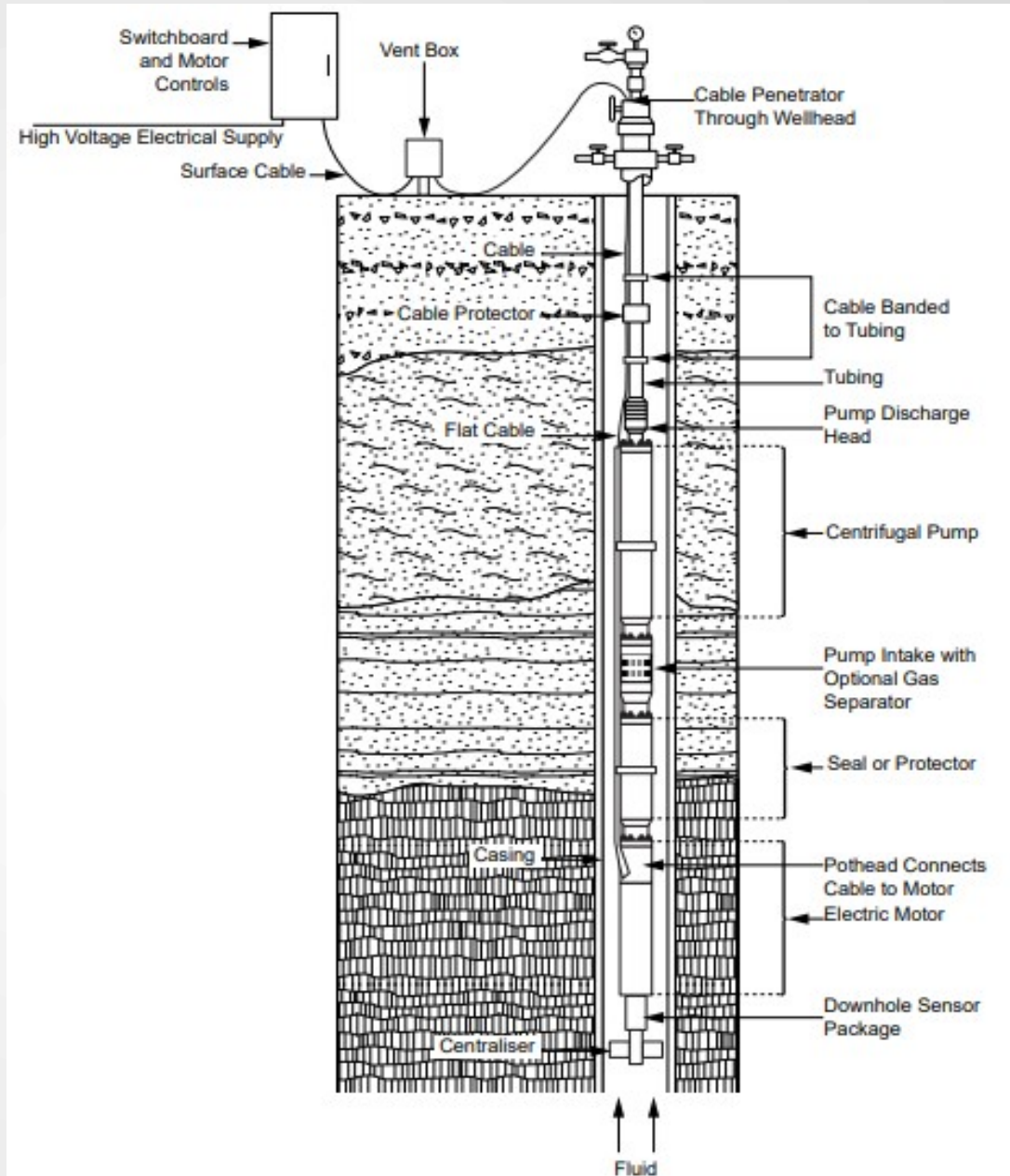
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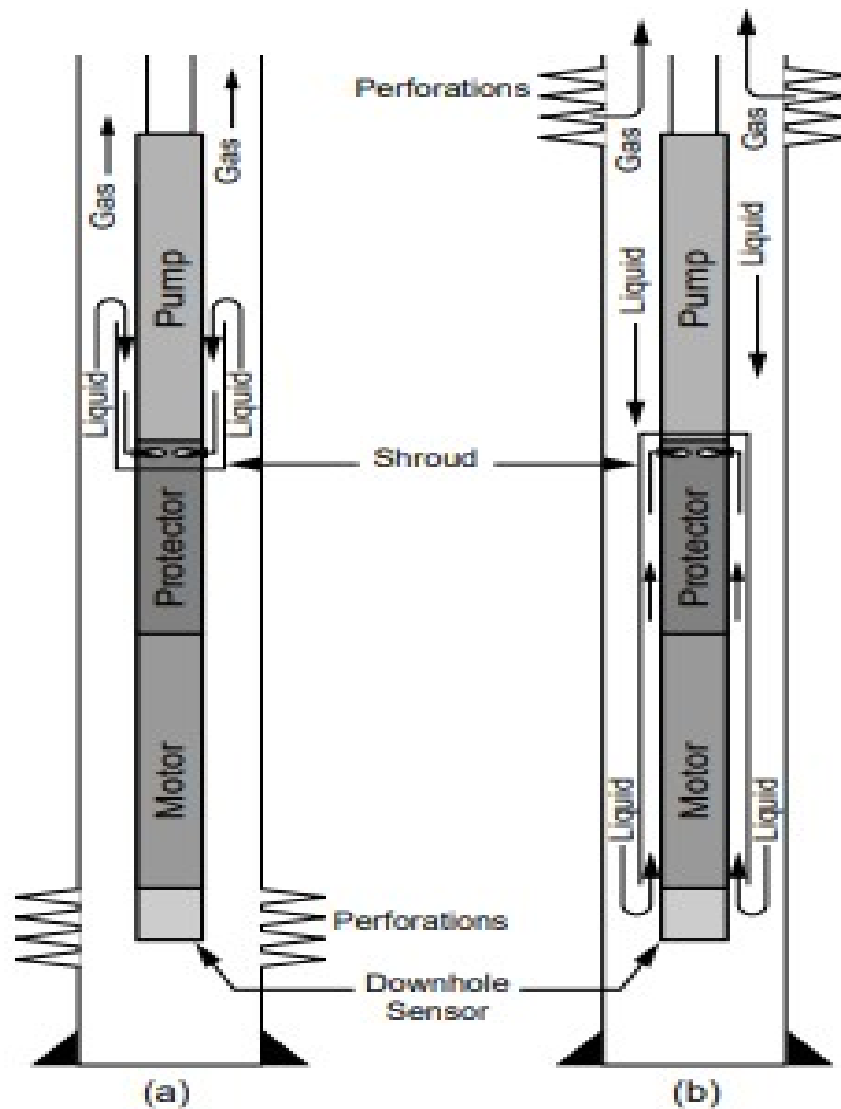
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Electric Submersible Pumps (ESP's) are a versatile form of artificial lift with pumps ranging from 150 to 60,000 bfpd in operation. A typical low pressure well that is being artificially lifted using an ESP system is illustrated in figure 17. The functions of the various components are summarised as follows:



- (a) Liquid/gas separation in the casing will only work when the upward velocity of the gas bubbles is greater than downward fluid velocity
- (b) The fitting of a shroud increases the maximum ESP diameter (requiring a large casing for installation of the same motor/pump combination or requires that a smaller diameter pump/motor be chosen)



TRANSFORMER

- A transformer is an electrical device used to step up the voltage for powering the ESP motor and pump in oil wells.



- •Voltage Transformation
- •Electrical Isolation
- •Voltage Regulation
- •Protection
- •Electrical Safety

MOTOR CONTROLLER

A motor controller is an electrical device used to regulate and control the operation of the ESP motor.



JUNCTION BOX

A junction box is an enclosure that houses electrical connections and components, facilitating the connection of power cables and control lines to the ESP system components.



CABLE VENTING BOX

A cable venting box is a device that provides pressure equalization and protection for the cables used to power the ESP system in oil wells. It helps prevent moisture and contaminants from entering the cable connections.

- Pressure Equalization
- Preventing Moisture Ingress
- Maintaining Cable Integrity
- Preventing Gas Lock
- Protection Against Environmental Factors
- Safety



WELLHEAD

a wellhead is the structure at the top of an oil well that contains the equipment and valves needed to control the flow of fluids into and out of the well.

- Support and Sealing
- Fluid Containment
- Cable and Tubing Entry
- Monitoring and Control
- Safety
- Maintenance Access
- Environmental Protection
- Well Shut-in Capability



Seal chamber section: located below the pump and above the motor.

Function : protect the motor oil.

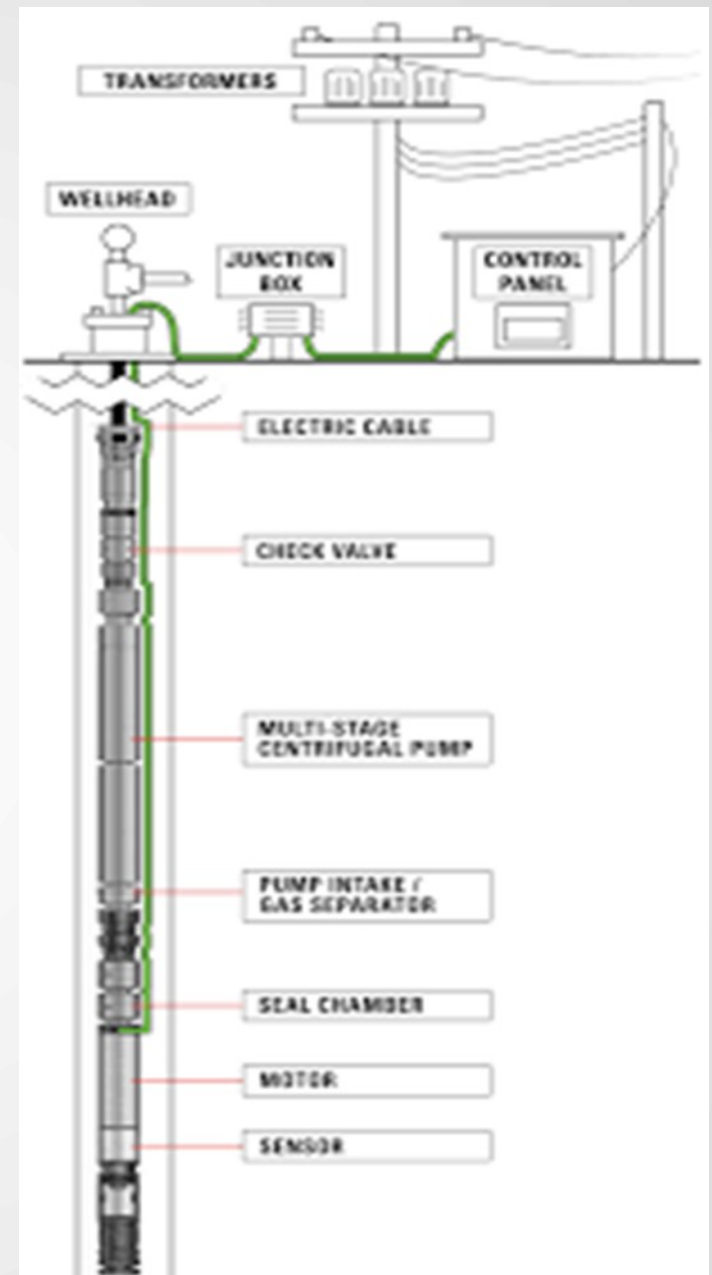
It allow pressure equalization.

ESP Motor: located below the seal section and above sensor.

Design of motor:

(induction motor)(two poles)(three phases)
(the speed 3600 RPM at 60 hertz)

- **Downhole sensor:** attached to the bottom of motor
- Function: monitoring of pressure and temperature
- Result send to power cable and then to the surface



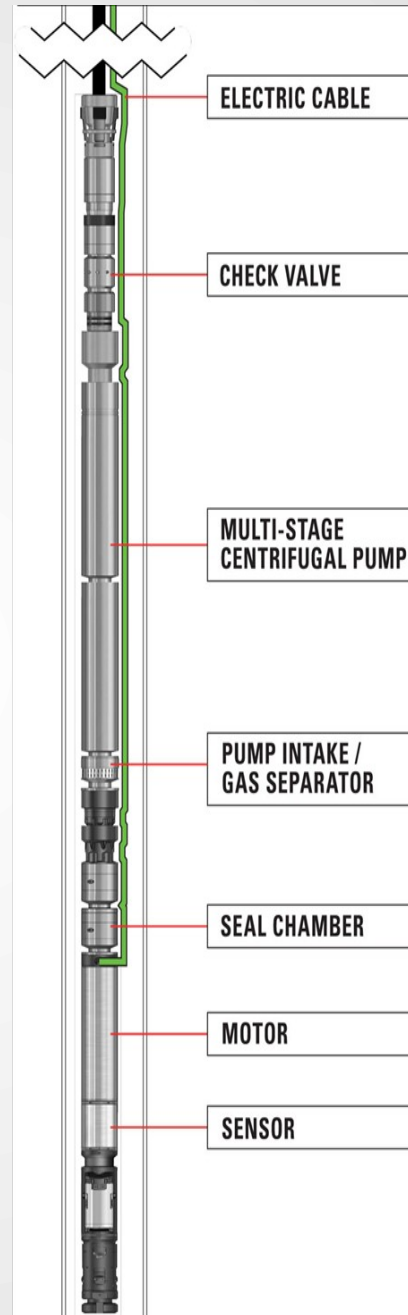
Check and drain valve:

Check valve use in the tubing string. Installed above two or three joint to keep the tubing full liquid when the pump is off.

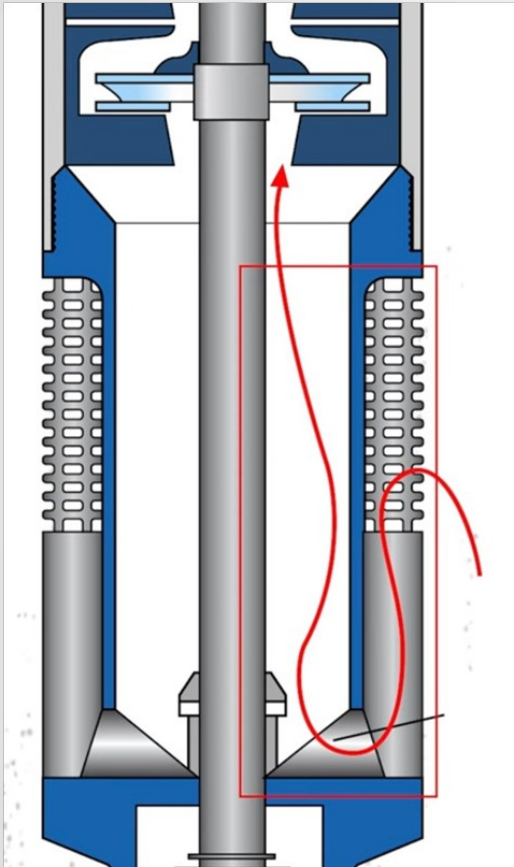
When check valve is used It is recommend to install drain valve, Located one joint above the check valve.



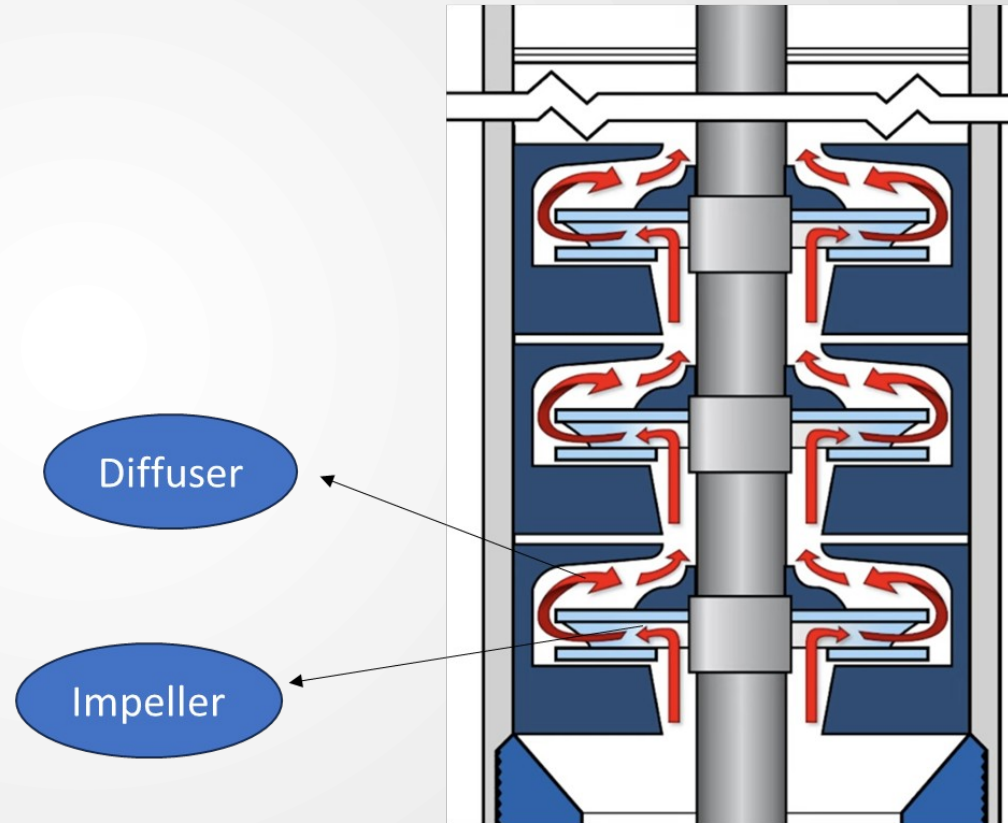
Subsurface equipment



Pump intake



Centrifugal pump



Y-tool and bypass system in ESP

- Y- tool system

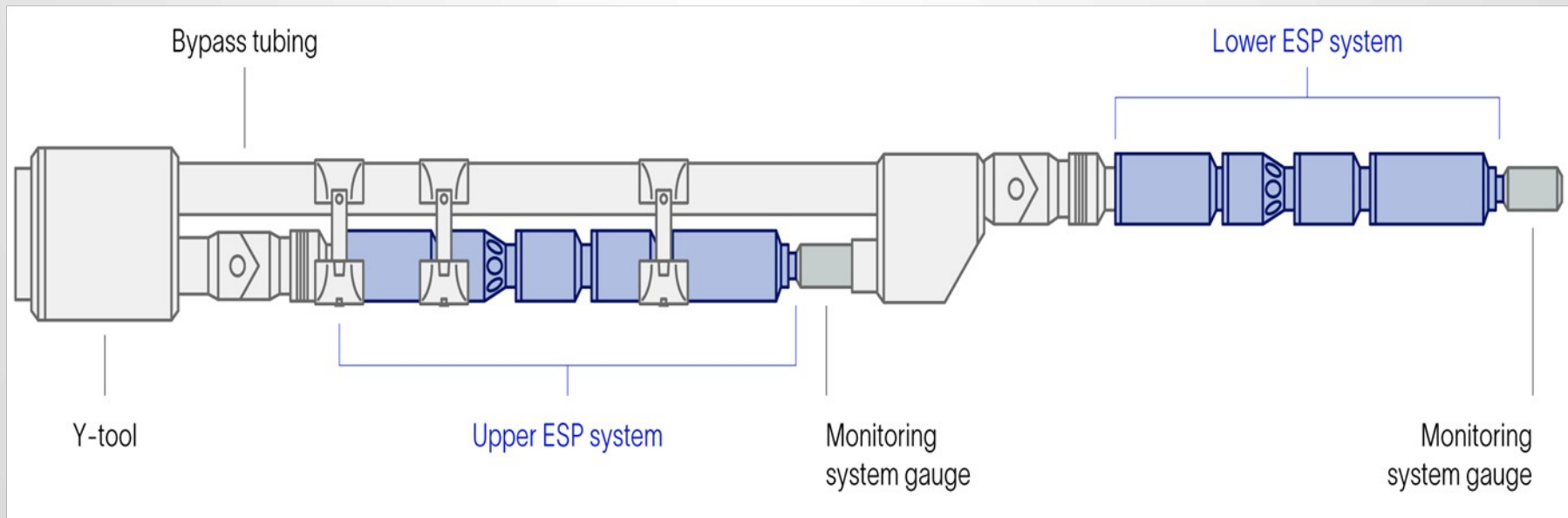
is a configuration of down hole components that includes the motor, pump stages, and electrical cable, designed to connect and optimize the operation of the ESP



Y-tool and bypass system in ESP

- Bypass system

The bypass system is primarily focused on diverting a portion of the produced fluid away from the pump to control pressure, temperature, and handle gas or solids



Advantages of (ESP) vs Disadvantages of (ESP)

- **Advantages of Electrical Submersible Pumps (ESP):**

1. **High Efficiency:** ESPs are known for their high efficiency in lifting fluids from wells, making them cost-effective for oil and water production.
2. **Variable Speed:** ESPs can be adjusted to varying flow rates and well conditions, providing operational flexibility.
3. **Deep Well Capability:** They are suitable for deep wells, even those several thousand feet deep, due to their submersible nature.
4. **Low Maintenance:** ESPs have fewer moving parts compared to other pump types, leading to lower maintenance requirements.

Disadvantages of Electrical Submersible Pumps (ESP):

1. **Cost:** ESPs can be expensive to purchase and install, making them less suitable for low-production or temporary wells.
2. **Electrical Requirements:** They demand a stable and reliable power supply, which can be a challenge in remote or offshore locations.
3. **Specialized Knowledge:** Proper design and maintenance of ESPs require expertise, which may not be readily available in all regions.
4. **Corrosion Issues:** In corrosive environments, the materials used in ESPs may degrade, necessitating corrosion-resistant designs.
5. **Size Limitations:** ESPs have size limitations, and for extremely high flow rates, multiple units may be needed.

Finally The choice to use an ESP depends on the specific well conditions and production requirements.

Thank you for listening

References

- ❑ [petroleume](#)
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